Enterprise PDF Processing

Searchable sample for extraction, layout, metadata, citation, and recursive chunking tests

Document Purpose

This PDF intentionally contains prominent titles, ordinary paragraphs, positioned content, a two-column layout, a structured table, repeated page footers, and long narrative text. It is designed to validate page extraction, visual reading order, title detection, text-block grouping, paragraph boundaries, metadata generation, citation creation, and token-aware splitting.

All text on the first four pages is embedded and selectable. A PDF extractor should report meaningful page text without OCR. The final page contains only a light geometric marker and no embedded natural-language text so that empty-page behavior can be verified.

> **Expected citation example: PdfSample 1.pdf - Page 1**

Validation Goals

• Extract page text in both stream order and displayed layout order.

• Retain page number, page dimensions, source file details, and block coordinates.

• Detect title-like text using font size, style, position, and spacing heuristics.

• Preserve every meaningful structure while treating the selected mode as a preferred boundary.

Architecture Overview

This page uses two visual columns. Layout-aware extraction should read the left column from top to bottom and then the right column from top to bottom, instead of interleaving unrelated lines.

Ingestion Layer

The ingestion layer receives PDF files, validates the source, records document metadata, and opens the document through the Syncfusion PDF library.

Extraction Layer

The extraction layer reads each page, captures text, and collects line, word, glyph, font, and bounding-rectangle information when the installed API supports those details.

Planning Layer

The planner preserves all meaningful content while using Page, Title, TextBlock, Paragraph, Table, or OcrBlock as the preferred semantic boundary.

Chunking Layer

Oversized structures fall back to smaller semantic boundaries. Tables split by row groups, paragraphs split by sentences, and lengthy text eventually uses word or token-safe fallback.

Reading Order Note

Basic extraction may follow the order of operators in the PDF content stream. Layout extraction should more closely match the visual order shown by a PDF reader. The probe should print both versions so differences are visible.

Deployment Validation Matrix

This page contains a table intended to test table detection, row-group splitting, repeated headers, and page-level citations.

Validation Area	Test Method	Expected Result	Required
Application startup	Review startup logs	No critical errors	Yes
Health endpoint	Send HTTP request	Successful response	Yes
Database connection	Execute health check	Connection available	Yes
Authentication	Request protected resource	Valid user is authorized	Yes
Document processing	Process sample PDF	Chunks and citations produced	Yes
Metrics	Query metrics endpoint	Current metrics returned	No

Table Splitting Guidance

If the complete table exceeds the token limit, the chunker should divide it between complete rows and repeat the header in every generated table part. A single oversized row may use a safe fallback, but random fragments from unrelated rows should not be combined.

Operational Guidance and Recursive Splitting

Deployment verification must confirm more than a successful process start. The application should respond through its health endpoint, connect to required dependencies, process a representative PDF, publish logs and metrics, and preserve sufficient diagnostic information for troubleshooting and rollback.

When a preferred PDF boundary exceeds the configured token limit, the engine should recursively use smaller meaningful boundaries. A page can fall back to detected titles or text blocks. A text block can fall back to paragraphs. A paragraph can fall back to sentences, then words, and finally a token-safe representation when no larger boundary fits.

Overlap should be applied only between adjacent parts created from the same original source structure. It should preserve context without causing the resulting chunk to exceed MaxTokens. Structured table syntax should not receive ordinary textual overlap because repeated headers already provide the required context.

Every final chunk should retain a deterministic document identifier, sequential chunk index, original block identifier, page number, content type, metadata, and citation. Repeated runs against the same unchanged PDF and options should produce stable identifiers suitable for vector database upsert and re-index operations.

Expected Fallback Order

• Preferred mode boundary

• Child semantic boundary

• Paragraph or block boundary

• Sentence boundary

• Word or token-safe fallback